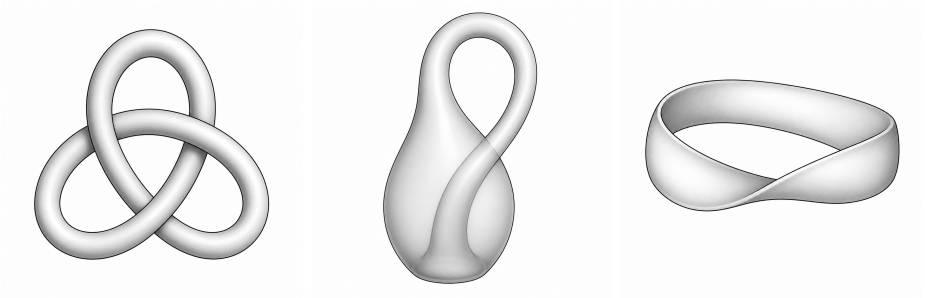
Topology

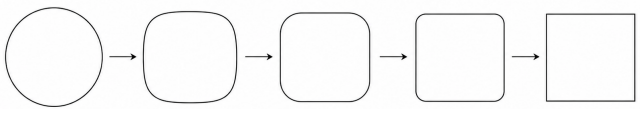
Students who have engaged with but not studied topology typically envision weird shapes, geometric objects and knots.



While these things all show up eventually, the definition of a topological space often stumps students when they first encounter it.

Topology looks at geometry but does so without measurement. The main thing the 'topology' of an object allows us to study is how it behaves when you stretch it. We call two objects *homeomorphic* when you can continuously deform one object into the other - i.e. without creating or filling holes, creating or losing dimensions, gluing itself or passing through itself. It turns out, all the concepts we care about in a strictly topological sense do not require actual measurement. When it comes to looking at holes, knots and self-intersection, the distance between two points doesn't matter. This is because it doesn't matter how big the hole is, we just care that it's there. The only interesting information we care about in topology is the underlying structure of the object, we don't care about about size or distance.

Consider a square and a circle - specifically the boundaries. A circle is defined as the collection of points equidistant from some chosen point, and a square defined as a quadrilateral with four equal length sides. Topologically, these are both the same, because they both have one hole, have an inside and an outside, and do not self intersect. So these two objects are *homeomorphic*, and when we picture a homeomorphism, we picture it like an animation of one object turning into another, as below



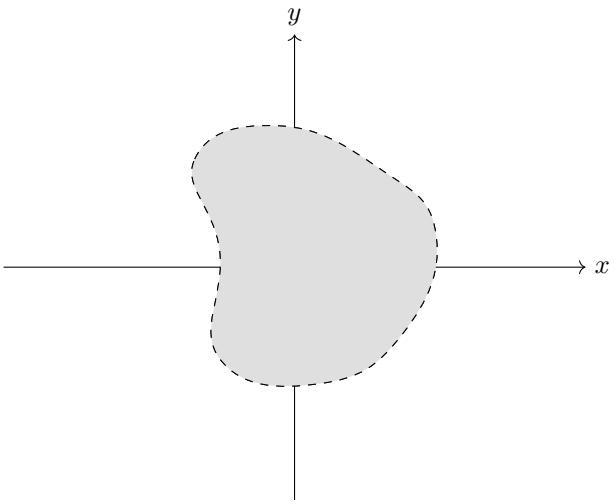
None of these mentioned topological concepts require the concept of a numerical distance.

A lot of the spaces so far in this book - vector spaces, metric spaces, groups, rings - are sets of objects with some sort of structure, whether that be addition, multiplication, scaling, or distance. A topological space has none of this - because it doesn't need it! A topological space is about as bare bones as it gets, basically just a set of points, with a small requirement on the structure so that we can examine closeness (but not numerically).

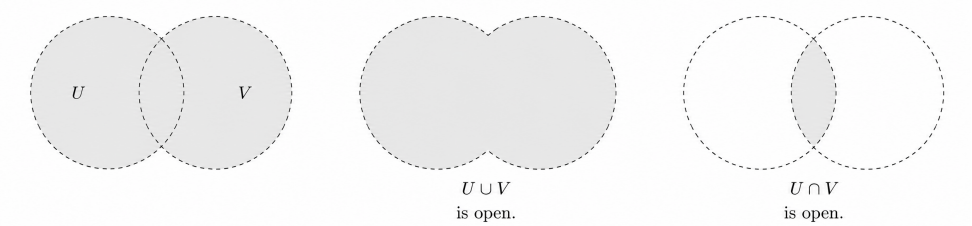
So what is a topological space? In order to understand, we need to look at one concept. Topologically, we still care about the boundaries of our objects, so we want to ensure our extra structure accounts for this. When we look at two sets U and V, there's two common operations we can perform

- Union ($U \cup V$): Returns all the values from U and V
- Intersection ($U \cap V$): Returns the values common to both U *and* V

An open set in $\mathbb{R}$ is an interval where the parentheses are round, e.g $(0, 1)$. Another way to think of open sets is that open sets do not have their boundaries as part of the set. If we were to draw a generic set in the $x-y$ plane, i.e. an open set of $\mathbb{R}^2$, we would represent an open set with a dashed line and a closed set with a solid line. So a diagram of a generic open set on the $x-y$ plane would look like



If we take a union of two open sets or an intersection of two open sets, they will both return open sets.

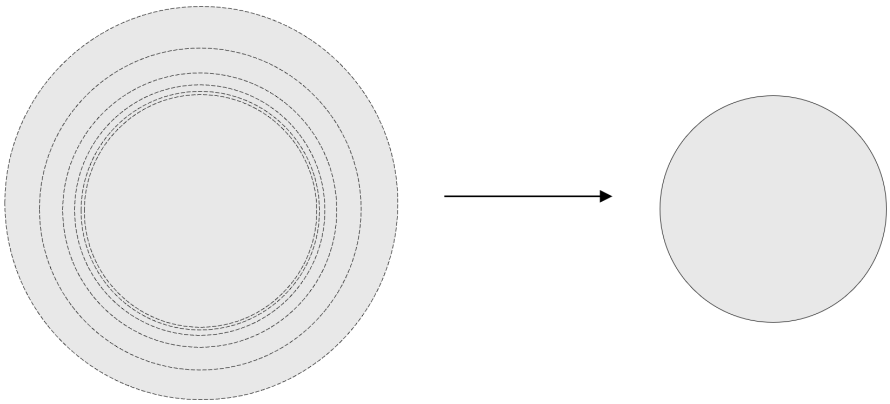


And this is true for a union and intersection to any number of finite open sets. It turns out even a union of an infinite series of sets is open. However, an intersection of an infinite set of open sets *may* be closed.

Consider a series of concentric open circles.

The radius of the first ball is $r_1 = 1 + \frac{1}{1} = 2$, and we'll call this set U_1.
The second circle will have a radius $r_2 = 1 + \frac{1}{2} = \frac{3}{2}$ and we'll call this set U_2.
Similarly, the third circle will have a radius $r_3 = 1 + \frac{1}{3} = 1\frac{1}{3}$ and be called U_3.

This infinite series of concentric circles converges in on a circle of radius $r = 1$.



So the intersection of all these sets consists of points that are inside all the sets U_n. This consists of all the points inside the circle of $r = 1$ including the boundary. In particular, this means the intersection is *not open*, it is *closed*.

Definition of a topological Space

So now that we know this, we can make sense of the definition of a topological space. A topological space is a set of points X, coupled with a *topology*, notated τ, (the greek letter *tau*), which is a list of sets that we declare to be *'open'*. In this list of open sets τ we must have:

- The empty set (notated $\emptyset$) and the whole set X

- For any finite or infinite combination of open sets, the union of these sets must also be in the list
- For any finite combination of open sets, the intersection of these sets must also be in the list.

Just like we enforce certain conditions on our distance function to align with our intuition in $\mathbb{R}^n$, we also enforce this condition on the list of open sets to align with our rules of union and intersection of open sets in $\mathbb{R}^n$.

So you can probably see why students get stumped when they have this definition given to them. The definition does not at all look conducive to studying topology as we know it to be. But it turns out this is all the structure we need to study holes, knots, boundaries, self intersection and deformations. But you may ask, why do we need to declare what sets are 'open'? This is to make continuity work. Continuity was discussed earlier, however there is an equivalent definition.